

THE SCOTS COLLEGE



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

YEAR 12

EXTENSION 2 MATHEMATICS

AUGUST 2001

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Exam continues over

TIME ALLOWED: **THREE HOURS**
[PLUS 5 MINUTES READING TIME]

OUTCOMES:

- Uses the relationship between algebraic and geometric representations of complex numbers and of conic sections. [E3]
- Uses efficient techniques for algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials. [E4]
- Combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions. [E6]
- Uses the techniques of slicing and cylindrical shells to determine volumes. Applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems. [E7]
- Applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems. [E8]
- Uses ideas and techniques from calculus to solve problems in mechanics involving resolution of forces, resisted motion and circular motion. [E5]

a. Find: [4 MARKS]

(i) $\int x^3 \log_e x \, dx$

(ii) $\int \sin^3 \theta \, d\theta$

b. Find the exact value of: [3 MARKS]

$$\int_1^7 \frac{dx}{x^2 - 8x + 25}$$

c. Using the substitution $u = \cos x$ to evaluate: [3 MARKS]

$$\int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\cos^2 x} \, dx$$

d. [5 MARKS]

(i) Show that $(1 - \sqrt{x})^{n-1} \sqrt{x} = (1 - \sqrt{x})^{n-1} - (1 - \sqrt{x})^n$

(ii) If $I_n = \int_0^1 (1 - \sqrt{x})^n \, dx$ for $n \geq 0$ show that $I_n = \frac{n}{n+2} I_{n-1}$ for $n \geq 1$

(iii) Deduce that $\frac{1}{I_n} = \binom{n+2}{n}$ for $n \geq 0$

a. Let $z = 3 - 2i$ and $u = -5 + 6i$ [4 MARKS]

(i) Find $\text{Im}(uz)$

(ii) Find $|u - z|$

(iii) Find $\overline{-2iz}$

(iv) Express $\frac{u}{z}$ in the form $a + ib$, where a and b are real numbers.

b. On separate Argand diagrams sketch: [4 MARKS]

(i) $\{z : |z - 2i| < 2\}$

(ii) $\{z : \arg(z - (1 + i)) = -\frac{3\pi}{4}\}$

c. z_1 and z_2 are two complex numbers such that $\frac{z_1 + z_2}{z_1 - z_2} = 2i$ [7 MARKS]

(i) On an Argand diagram show vectors representing: z_1 , z_2 , $z_1 + z_2$ and $z_1 - z_2$.

(ii) Show that $|z_1| = |z_2|$

(iii) If α is the angle between the vectors representing z_1 and z_2 , show that $\tan \frac{\alpha}{2} = \frac{1}{2}$

(vi) Show that $z_2 = \frac{1}{5}(3 + 4i)z_1$

QUESTION THREE

[START A NEW ANSWER BOOKLET]

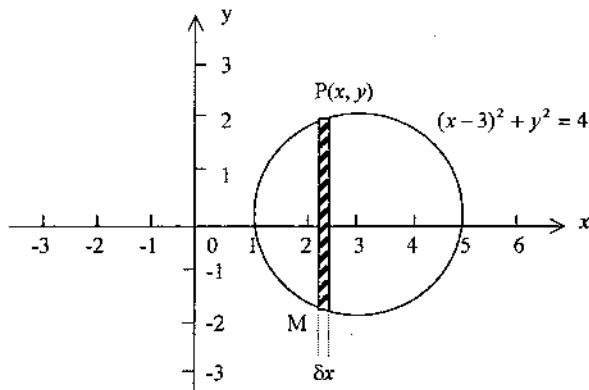
- a. The base of a solid is the region between the lines $y = 3x$ and $y = -x$ from $x = 0$ to $x = 2$. Each cross section by planes perpendicular to the x -axis is a square with its side determined by the base. Calculate the volume of the solid. [3 MARKS]

- b. The area bounded by the curve $y = x^2 + 1$ and the line $y = 3 - x$ is rotated about the x -axis. [4 MARKS]

- (i) Sketch the curve and the line clearly showing and labelling all the points of intersection.
- (ii) By considering slices perpendicular to the x -axis, find the volume of the solid formed.

- c. The graph below is of the circle $(x - 3)^2 + y^2 = 4$. [8 MARKS]

$P(x, y)$ is a point on the circumference of the circle. PM is the left-hand end of a strip of width δx which is parallel to the y -axis.



- (i) Show, using the method of cylindrical shells, that the volume V of the doughnut-shaped solid formed when the region inside the circle is rotated about the y -axis is given by:

$$V = 4\pi \int_1^5 x\sqrt{4 - (x-3)^2} dx$$

- (ii) Hence, by using the substitution $u = x - 3$ or otherwise find the volume of the doughnut.

QUESTION FOUR

[START A NEW ANSWER BOOKLET]

Consider the function $f(x) = x - 2\sqrt{x}$

[15 MARKS]

- a. Determine the domain of $f(x)$.
- b. Find the x intercepts of the graph of $y = f(x)$.
- c. Show that the curve $y = f(x)$ is concave upwards for all positive values of x .
- d. Find the coordinates of the turning point and determine its nature.
- e. Sketch the graph of $y = f(x)$ clearly showing all essential details.

- f. Hence, sketch on separate diagrams:

(i) $y = |f(x)|$

(ii) $y = f(x - 1)$

(iii) $y = f(|x|)$

(iv) $|y| = f(x)$

(v) $y = \frac{1}{f(x)}$

- a. Given that $z = -1 + \sqrt{3}i$ is a root of the equation $z^4 - 4z^2 - 16z - 16 = 0$, find the other roots.
[4 MARKS]

- b. Given that α, β and γ are the roots of the cubic equation $x^3 - x^2 + 5x - 3 = 0$, find:
[5 MARKS]

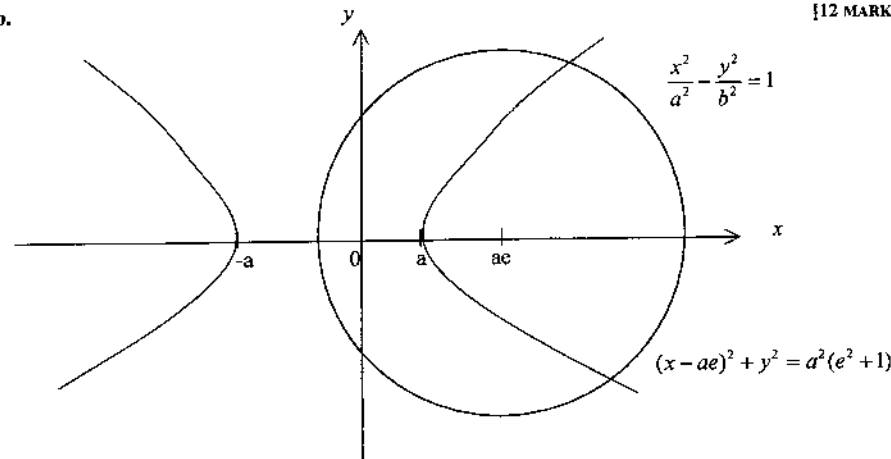
(i) the equation whose roots are $-\alpha, -\beta, -\gamma$.

(ii) the equation whose roots are $\alpha\beta, \alpha\gamma, \beta\gamma$.

- c. For what values of m does the equation $x^3 - 12x^2 + 45x - m = 0$ have three distinct solutions?
[6 MARKS]

- a. A hyperbola has asymptotes $y = x$ and $y = -x$. It passes through the point $(3, 2)$. Find the equation of the hyperbola and determine its eccentricity and foci.
[3 MARKS]

- b. [12 MARKS]



- (i) Show that the tangent at $P(a \sec \theta, b \tan \theta)$ on the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

has equation

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} - 1 = 0$$

- (ii) Show that if the tangent at P is also tangent to the circle with centre $(ae, 0)$ and radius $a\sqrt{e^2 + 1}$, then show $\sec \theta = -e$.
- (iii) Given that $\sec \theta = -e$, deduce that the points of contact P, Q on the hyperbola of the common tangents to the circle and hyperbola are the extremities of a latus rectum of the hyperbola, and state the coordinates of P and Q .
- (iv) Find the equations of the common tangents to the circle and hyperbola, and find the coordinates of their points of contact with the circle.

- a. A mass of 10kg falls freely from rest through 10 metres and then comes to rest again after penetrating 0.2 metres of sand.

Find the resistance of the sand, assumed constant.

[4 MARKS]

- b. A particle moving in a straight line experiences a force numerically equal to $\left(x + \frac{1}{x}\right)$ newtons per unit mass, towards the origin. The particle starts from rest, d units from the origin.

[4 MARKS]

(i) Find an expression for its speed in terms of x .

(ii) Hence or otherwise, deduce its speed when it is half way from the origin.

- c. An object of irregular shape and of mass 100kg is found to experience a resistive force, in newtons, of magnitude one-tenth the square of its velocity in metres per second when it moves through air [use $g = 9.8 \text{ ms}^{-2}$].

[7 MARKS]

If the object falls from rest under gravity:

(i) show that acceleration is given by $a = g - \frac{v^2}{1000}$.

(ii) calculate its terminal velocity.

(iii) calculate the maximum height, to the nearest metre, of the release point above the ground, if the object attains a speed of 80% of its terminal velocity before striking the ground.

- a. Let α , β and γ be the roots of the cubic equation $x^3 + Ax^2 + Bx + 8 = 0$, where A , and B are real. Furthermore $\alpha^2 + \beta^2 = 0$ and $\beta^2 + \gamma^2 = 0$.

[5 MARKS]

(i) Explain why β is real and α and γ are not real.

(ii) Show that α and γ are purely imaginary.

(iii) Find A and B .

- b. It is given that if $J_n = \int \cos^{n-1} x \sin nx \, dx$ and $n \geq 1$ then:

[5 MARKS]

$$J_n = \frac{1}{2n-1} [(n-1)J_{n-1} - \cos^{n-1} x \cos nx]$$

Use this reduction formula to show that:

$$\int_0^{\frac{\pi}{2}} \cos^2 x \sin 3x \, dx = \frac{1}{60} (28 - \sqrt{2})$$

c.

(i) Prove that $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = 2 \sec^n \theta \cos n\theta$

[5 MARKS]

(ii) Hence prove that $\text{Re}(1 + i \tan \frac{\pi}{8})^8 = 64(17 - 12\sqrt{2})$.

Ext 2 Trial Exam August 2001

Question 1

$$\begin{aligned} a) \int x^3 \ln x \, dx \\ = \ln x \cdot \frac{x^4}{4} - \frac{1}{4} \int x^3 \, dx \\ = \frac{x^4}{4} \ln x - \frac{x^4}{16} + C \end{aligned}$$

$$\begin{aligned} b) \int \sin^3 \theta \, d\theta &= \int (1 - \cos^2 \theta) \sin \theta \, d\theta \\ &= \int \sin \theta - \cos^2 \theta \sin \theta \, d\theta \\ &= -\cos \theta + \frac{1}{3} \cos^3 \theta + C \end{aligned}$$

$$\begin{aligned} c) \int_4^7 \frac{dx}{x^2 - 8x + 25} &= \int_4^7 \frac{dx}{(x-4)^2 + 9} \\ &= \frac{1}{3} \left[\tan^{-1} \left(\frac{x-4}{3} \right) \right]_4^7 \\ &= \frac{1}{3} \left[\tan^{-1}(1) - \tan^{-1}(0) \right] \\ &= \frac{\pi}{12} \end{aligned}$$

$$\begin{aligned} c) \quad u &= \cos x \\ \frac{du}{dx} &= -\sin x \\ x=0, \quad u &= 1 \\ x=\frac{\pi}{3}, \quad u &= \frac{1}{2} \\ I &= \int_1^{\frac{1}{2}} \frac{(1 - \cos^2 x) \sin x}{\cos^2 x} \, dx \\ &= - \int_{\frac{1}{2}}^1 \frac{(1-u^2) \cdot du}{u^2} \\ &= - \left[-\frac{1}{u} - u \right]_{\frac{1}{2}}^1 \\ &= -1 - 1 - \left(-2 - \frac{1}{2} \right) \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} a) \quad z &= 3-2i, \quad u = -5+6i \\ (i) \quad uz &= (3-2i)(-5+6i) \\ &= -15 + 18i + 10i + 12 \\ &= -3 + 28i \end{aligned}$$

$$\therefore \operatorname{Im}(uz) = 28$$

$$\begin{aligned} (ii) \quad u-z &= -5+6i - (3-2i) \\ &= -8+8i \\ |u-z| &= \sqrt{64+64} \\ &= \sqrt{128} \\ &= 8\sqrt{2} \end{aligned}$$

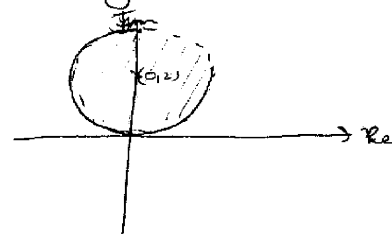
$$b) \quad |z-2i| < 2$$

$$\text{Let } z = x+iy$$

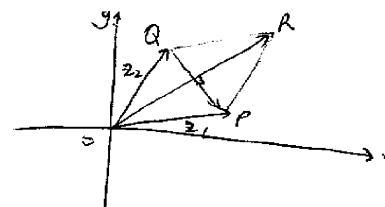
$$\text{Consider } |x+i(y-2)| < 2$$

$$\sqrt{x^2 + (y-2)^2} < 2$$

$$x^2 + (y-2)^2 < 4$$



$$c) \quad \frac{z_1 + z_2}{z_1 - z_2} = 2i$$



$$\begin{aligned} (iii) \quad -2iz &= -2i(3-2i) \\ &= -4-6i \end{aligned}$$

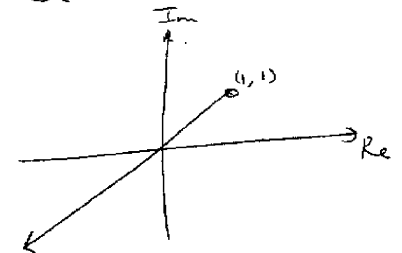
$$\overline{-2iz} = -4+6i$$

$$\begin{aligned} (iv) \quad \frac{-5+6i}{3-2i} \times \frac{3+2i}{3+2i} \\ = \frac{-15-10i+18i-12}{9+4} \\ = \frac{-27+8i}{13} \end{aligned}$$

$$(ii) \quad \arg(z - (1+i)) = \frac{3\pi}{4}$$

$$\text{Let } z = x+iy$$

$$\arg(x-1+i(y-1)) = \frac{3\pi}{4}$$



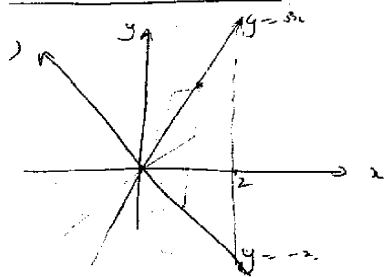
$$\vec{OP} = z_1, \quad \vec{OQ} = z_2$$

$$\vec{OR} = z_1 + z_2$$

$$\vec{OP} = z_1 - z_2$$

OPQR is parallelogram

Question Three



$$\text{length} = 3x + (4x)$$

$$= 7x$$

$$SA = 16x^2$$

$$\delta V = 16x^2 \delta x$$

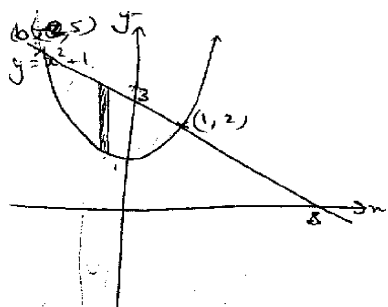
$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^2 16x^2 \delta x$$

$$V = \int_0^2 16x^2 dx$$

$$= \left[\frac{16x^3}{3} \right]_0^2$$

$$= \frac{128}{3} \text{ cubic units.}$$

i) (1)



$$x^2 + 1 = 3 - x$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$x = -2 \text{ or } x = 1$$

$$y = 5 \text{ or } y = 2$$

$$\text{ii) } y_1 = 3 - x \quad y_2 = x^2 + 1$$

$$SA = \pi((3-x)^2 - (x^2+1)^2)$$

$$\delta V = \pi(8 - 6x + x^2 - (x^4 + 2x^2 + 1)) \delta x$$

$$= \pi(8 - 6x + x^2 - x^4 - 2x^2 - 1) \delta x$$

$$= \pi(8 - 6x - x^2 - x^4) \delta x$$

$$V = \pi \int_{-2}^1 8 - 6x - x^2 - x^4 dx$$

$$= \pi \left[8x - 3x^2 - \frac{x^3}{3} - \frac{x^5}{5} \right]_{-2}^1$$

$$\therefore V = \pi \left(\frac{67}{15} - \left(-\frac{284}{15} \right) \right)$$

$$= \frac{117\pi}{5} \text{ cubic units.}$$

Question 4

$$f(x) = x - 2\sqrt{x}$$

$$a) \text{ Domain: } x \geq 0$$

$$b) \text{ Sub } f(x) = 0, \quad x - 2\sqrt{x} = 0$$

$$\sqrt{x}(x^{\frac{1}{2}} - 2) = 0$$

$$\therefore x = 0 \text{ or } x = 4$$

$$c) f'(x) = 1 - 2 \cdot \frac{1}{2} x^{-\frac{1}{2}}$$

$$= 1 - \frac{1}{\sqrt{x}}$$

$$f''(x) = -\frac{1}{2} \cdot -x^{-\frac{3}{2}}$$

$$= \frac{2}{x^{\frac{3}{2}}}$$

$$= \frac{1}{2\sqrt{x^3}} \text{ for } x \geq 0.$$

$$\therefore f''(x) > 0 \text{ for } x > 0 \therefore y = f(x) \text{ concave up for } x > 0.$$

$$d) f'(x) = 1 - \frac{1}{\sqrt{x}}$$

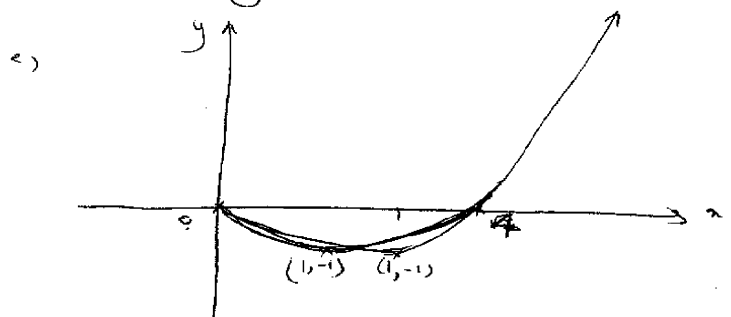
$$\text{for stat pts, } f'(x) = 0.$$

$$1 - \frac{1}{\sqrt{x}} = 0$$

$$\sqrt{x} = 1$$

$$x = 1.$$

$$\text{When } x = 1, y = -1, f''(x) > 0 \therefore (1, -1) \text{ min turning}$$



Question 5

$$z_1 = -1 + \sqrt{3}i \quad \therefore z_2 = -1 - \sqrt{3}i$$

$$(z - z_1)(z - z_2) = z^2 - (z_1 + z_2)z + z_1 z_2$$

$$= z^2 + 2z + 4$$

$$z^4 - 4z^2 - 16z - 16 = (z^2 + 2z + 4)(z^2 + Az + B)$$

$$= (z^2 + 2z + 4)(z^2 + Az - 4)$$

$$\text{coeff of } z^2, -4 = -4 + 2A + 4$$

$$\therefore A = -2$$

$$\therefore \text{factor } z^2 - 2z - 4$$

$$z = \frac{2 \pm \sqrt{4 - 4(1)(-4)}}{2(1)}$$

$$z = \frac{2 \pm \sqrt{20}}{2}$$

$$z = 1 \pm \sqrt{5}$$

$$x^3 - x^2 + 5x - 3 = 0$$

$$x = -x \quad \therefore x = -x$$

$$(-x)^3 - (-x)^2 + 5(-x) - 3 = 0$$

$$-x^3 - x^2 - 5x - 3 = 0$$

$$x^3 + x^2 + 5x + 3 = 0$$

$$x = 3$$

$$x = \frac{3}{x} \quad \therefore x = \frac{3}{x} \quad \therefore x = \frac{3}{x}$$

$$x = \frac{3}{x}$$

$$x = \frac{3}{x}$$

$$\left(\frac{3}{x}\right)^3 - \left(\frac{3}{x}\right)^2 + 5\left(\frac{3}{x}\right) - 3 = 0$$

$$\frac{27}{x^3} - \frac{9}{x^2} + \frac{15}{x} - 3 = 0$$

$$27 - 9x + 15x^2 - 3x^3 = 0$$

$$3x^3 - 15x^2 + 9x - 27 = 0$$

$$x^3 - 5x^2 + 3x - 9 = 0$$

a) Rectangular hyperbola of form

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad e = \sqrt{2}$$

$$\text{Sub } (3, 2) \quad a^2 = 5 \quad \therefore \text{foci } (\pm ae, 0)$$

$$\therefore x^2 - y^2 = 5 \quad (\pm \sqrt{10}, 0)$$

$$b) \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad P(a \sec \theta, b \tan \theta)$$

$$\frac{2x}{a^2} - \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{b^2 x}{a^2 y}$$

$$\therefore \text{At } P, \quad \frac{dy}{dx} = \frac{b^2 a \sec \theta}{a^2 b \tan \theta}$$

$$= \frac{b \sec \theta}{a \tan \theta}$$

Let the equation of the tangent be

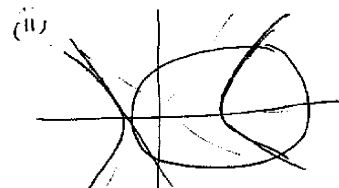
$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$a \tan \theta y - ab \tan^2 \theta = b \sec \theta x - ab \sec^2 \theta$$

$$b \sec \theta x - a \tan \theta y = ab(\sec^2 \theta - \tan^2 \theta)$$

$$b \sec \theta x - a \tan \theta y = ab$$

$$\therefore ab \frac{\sec \theta x}{a} - \frac{\tan \theta y}{b} = 1$$



Perpendicular Dist from C to tangent.

$$\therefore \text{dist} = \frac{|-a \sec \theta + 0 - 1|}{\sqrt{\frac{a^2 \sec^2 \theta}{a^2} + \frac{\tan^2 \theta}{b^2}}}$$

If $\sec \theta = -e$ then $e^2 = \tan^2 \theta + 1$
 $\tan \theta = \pm \sqrt{e^2 - 1}$

Coordinates of P & Q are $(a \sec \theta, b \tan \theta)$

& $(-ae, \pm b\sqrt{e^2 - 1})$ which lie on the latus

rectum $x = -ae$.

Eqn of tangent $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$.

Common tangents

$$-\frac{x e}{a} - \frac{y(\pm \sqrt{e^2 - 1})}{b} = 1$$

$$-x e - y(\pm \frac{a \sqrt{e^2 - 1}}{b}) = a$$

$$x e \pm y + a = 0.$$

$$\pm y = -(a + x e) \quad \text{--- (1)}$$

$$\pm y = \sqrt{a^2(e^2 + 1) - (x - ae)^2}$$

$$(a + x e)^2 = a^2(e^2 + 1) - (x - ae)^2$$

$$a^2 + 2axe + x^2 e^2 = a^2/e^2 + a^2 - x^2 + 2xae - a^2/e^2$$

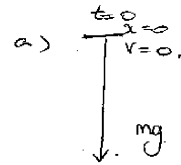
$$x^2(e^2 - 1) = 0$$

$$\therefore x = 0.$$

when $x = 0$, $y = \pm a$

pt of contact $(0, \pm a)$

Question 7



$$F = ma$$

$$10a = mg$$

$$a = g.$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = g$$

$$\therefore \frac{1}{2} v^2 = gx + c_1$$

$x=0, v=0, \therefore c_1=0$ when $x=10, v^2=20g$.

$$\therefore v^2 = 2gx$$

When it hits the sand $a = g - r$ (force of sand)

$$\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = (g - r)$$

$$\frac{1}{2} v^2 = (g - r)x + c_2$$

when $x=0, v^2=20g$.

$$10g = c_2$$

$$v^2 = 2(g - r)x + 20g.$$

$x = 0.2, v = 0.$

$$0 = 0.4(g - r) + 20g.$$

$$g - r = \frac{-20g}{0.4}$$

$$r = g + \frac{20g}{0.4}$$

$$r = 51g \text{ m/s}^2$$

$\therefore$ resistance = $510g$ Newtons.

$$\frac{dv}{dx} = \frac{1000g - v^2}{1000v}$$

$$\frac{dx}{dv} = \frac{1000v}{1000g - v^2}$$

$$x = 1000 \cdot \frac{-1}{2} \ln(1000g - v^2) + C_1$$

$$x = -500 \ln(1000g - v^2) + C_1$$

$$x=0, v=0, \therefore C_1 = 500 \ln(1000g)$$

$$\therefore x = -500 \ln\left(\frac{1000g - v^2}{1000g}\right)$$

$$\frac{-x}{500} = \ln\left(\frac{1000g - v^2}{1000g}\right)$$

$$e^{\frac{-x}{500}} = \frac{1000g - v^2}{1000g}$$

$$v^2 = 1000g(1 - e^{-x/500})$$

$$\text{then } x \rightarrow \infty, e^{-x/500} \rightarrow 0, \therefore v^2 \rightarrow 1000g$$

$$\therefore \text{terminal velocity} = 99 \text{ m/s.}$$

$$\text{Sub } v = 0.8 \sqrt{1000g}$$

$$\therefore x = -500 \ln\left(\frac{1000g - 640g}{1000g}\right)$$

$$x = -500 \ln\left(\frac{9}{25}\right)$$

$$x = 510.8 \text{ m}$$

Question 8

$$a) x^3 + Ax^2 + Bx + C = 0 \quad \alpha^2 + \beta^2 = 0 \quad \alpha + \beta + \gamma = 0$$

$$i) \text{ Since } \alpha^2 + \beta^2 = 0$$

$$\beta^2 = -\alpha^2 < 0$$

(if α were real, $\alpha^2 \geq 0$)

$\therefore \alpha$ is not real or similarly β is not real.

at least one of α or β is not real.

$$\text{Since } \alpha^2 + \beta^2 = \beta^2 + \gamma^2$$

$$\therefore \alpha^2 = \gamma^2$$

And since 3 roots exist - 1 real & 2 complex conjugate roots

$\therefore \alpha$ & γ are complex & β is real.

$$ii) \alpha^2 + \beta^2 = 0$$

$$\alpha^2 = -\beta^2$$

$$\alpha = -i\beta \quad \text{but } \beta \in \mathbb{R} \quad \therefore \alpha \text{ is purely imaginary}$$

$$\text{& } \alpha = \pm \gamma \quad \therefore \gamma = \mp i\beta \quad \therefore \gamma \text{ is purely imaginary}$$

$$iii) \text{ let roots be } i\beta, \beta, -i\beta$$

$$\text{product of roots } \beta^3 = -8$$

$$\beta = -2$$

$$\therefore \text{roots } -2i, -2, 2i$$

$$\therefore \text{sum of roots } -A = 2 \quad \therefore A = -2$$

$$\sum \alpha\beta \quad B = -4i + 4 + 4i \quad \therefore B = 4$$

$$\begin{aligned}
 \therefore \text{LHS} &= (1 + i \tan \theta)^n + (1 - i \tan \theta)^n \\
 &= \left(1 + i \frac{\sin \theta}{\cos \theta}\right)^n + \left(1 - i \frac{\sin \theta}{\cos \theta}\right)^n \\
 &= \frac{1}{\cos^n \theta} \left[(\cos \theta + i \sin \theta)^n + (\cos \theta - i \sin \theta)^n \right] \\
 &= \sec^n \theta (\cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta) \\
 &= 2 \sec^n \theta \cos n\theta \\
 &= \text{RHS}
 \end{aligned}$$

$$1) \operatorname{Re}(z) = \frac{1}{2} \operatorname{Re}(z + \bar{z})$$

$$\operatorname{Re}\left(1 + i \tan \frac{\pi}{8}\right)^8 = \frac{1}{2} \sec^8 \frac{\pi}{8} \cos \pi$$

$$= -\sec^8 \frac{\pi}{8}$$

$$2 \cos^2 \theta = \cos 2\theta + 1$$

$$\cos^2 \frac{\pi}{8} = \frac{1}{2} (\cos \frac{\pi}{4} + 1)$$

$$= \frac{1}{2} \left(\frac{\sqrt{2} + 1}{\sqrt{2}} \right)$$

$$\cos^8 \frac{\pi}{8} = \left(\frac{\sqrt{2} + 1}{2\sqrt{2}} \right)^4$$

$$\sec^8 \frac{\pi}{8} = \left(\frac{2\sqrt{2}}{\sqrt{2} + 1} \right)^4$$

$$= \frac{64}{17 + 12\sqrt{2}} \times \frac{17 - 12\sqrt{2}}{17 - 12\sqrt{2}}$$

$$= 64(17 - 12\sqrt{2})$$

$$\therefore -\sec^8 \frac{\pi}{8} = 64(12\sqrt{2} - 17)$$